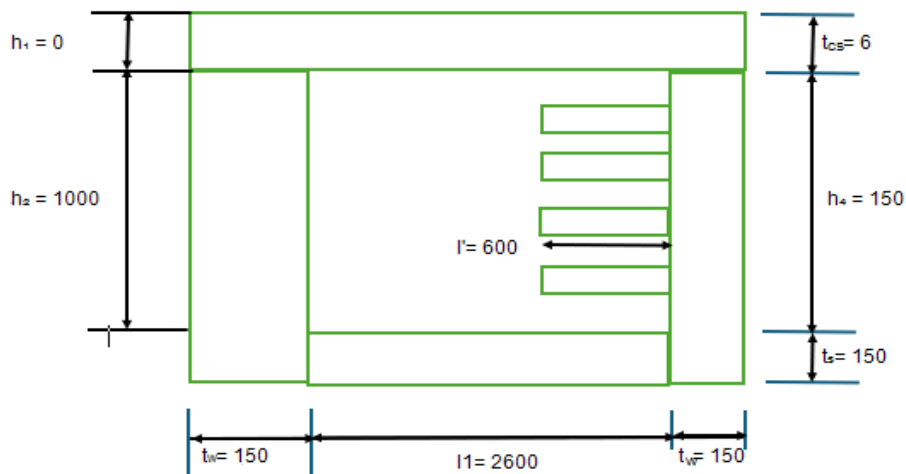


PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

1) Soil Parameters

Density of soil(Submerged Condition)	γ_{soil}	=	15.5	kN/m ³
Safe Bearing Capacity at foundation level	S.B.C	=	67.6	kN/m ²



All dimensions are in mm

Depth of water from F.G.L	h_w	=	5	m
Angle of internal friction of soil	ϕ	=	24	deg
Coefficient of Active Earth pressure	k_a	=	0.42	
Density of water	γ_w	=	10	kN/m ³

2) Strength Parameters

Unit Weight of Concrete	γ_{conc}	=	24.5	kN/m ³
Characteristic Strength of Concrete	f_{ck}	=	30	N/mm ²
Yield Strength of Steel	f_y	=	250	N/mm ²

3) Concrete Cover

Base Slab Top	=	50	mm
Base Slab Bottom	=	50	mm
Side wall Cover	=	75	mm
Cover slab	=	50	mm

4) Loading Parameters

Weight of cables per tray	w_c	=	0.75	kN/m
Weight of cable supports including trays		=	0.15	kN/m
Total Weight of Cables		=	0.90	kN/m



PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

Calculation of Effective span:

l+t	=	2600 + 6	
	=	2606	mm
C/C span of supports	=	2600 + 150	
	=	2750	mm
Hence effective span considered for design	=	2606	mm
U.D.L due to self weight of slab per width	=	0 x 0.15	
	=	0.00	kN/m
Live load on slab per width	=	1.5 x 0	
	=	0	kN/m
Concentrated load at mid-span of slab	=	0.75	kN
Moment due to self weight	=	0.00 x 2.606 ² / 8 + 0.75 x 2.606 / 4	
	=	0.49	kNm
Moment due to live load	=	0 x 2.606 ² / 8	
	=	0.00	kNm

Calculation for effective width of cover slab as per cl.24.3.2.a of IS 456-2000:

bef	=	k x (1 -x / lef)+a	
Ratio of l / lef	=	width of slab / effective span	
	=	0.00	
Value of constant 'k' from Table 14, IS:456	=	1.000	
Distance of conc. Load from support 'x'	=	1303	mm
Width of contact area of conc. Load 'a'	=	0	mm
bef	=	1.000 x 1303 x (1- 1303 / 2606)+ 0	
	=	651.50	mm
Actual width	=	0	mm
Hence eff width (lesser of bef & actual width)	=	0	mm
U.D.L due to self weight of slab	=	0.006 x 0 x 24.5	
	=	0.00	kN/m
Max Bending Moment	=	0.00 x 2.606 ² / 8 + 1.5 x 2.606 ² / 8 + 0.75 x 2.606 / 4	
	=	0.49	kNm
Total factored Bending Moment	=	0.73	kNm
Mu / bd ²	=	Infinity	N/mm ²
From Table 4, SP 16			
Percentage of steel required	pt	=	NaN %
Area of steel required		=	NaN mm ²



PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

Therefore, provide 0 Nos.of 0mm RTS as main reinforcement.

$$\begin{aligned} \text{Area of steel provided} &= 0.00 \\ &= 0.00 > \text{NaN mm}^2 \end{aligned}$$

Unsafe

Distribution steel :

$$\begin{aligned} \text{Min. area of steel required} &= 0.12 * 6 * (2600 + 150 + 150) / 100 \\ &= 21 \text{ mm}^2 \end{aligned}$$

$$\text{Considering a reinforcement spacing of} = 0 \text{ mm}$$

Therefore, provide 0 mm RTS as Distribution reinforcement.

$$\text{Area of steel provided} = \text{Infinity} < 21 \text{ mm}^2$$

Hence Safe

Check for deflection :

$$\text{Actual reinforcement steel \% provided 'pt'} = \text{NaN} \%$$

$$\text{Actual ratio of span / eff. depth} = -59.2$$

As per IS 456

$$\begin{aligned} \text{Steel Stress of service loads fs} &= 0.58 * f_y * A_{st \text{ req}} / A_{st \text{ prov}} \\ &= 0.00 \text{ N/mm}^2 \end{aligned}$$

$$\text{Modification factor from fig 4 : IS 456} = 0$$

$$\text{Clause 23.2.1 (a):IS 456,} = 0$$

For simply supported lx/d

$$\text{Modified ratio of lx/d} = 0$$

$$\text{Actual ratio of span / eff. depth} = -59.2 < 0$$

Hence Safe

Check for Shear:

Max. shear force :

$$\text{Shear due to selfweight (} W_1 l_2 / 2 \text{)} = 0.00 \times 2.606 / 2$$

$$= 0.000 \text{ kN}$$

$$\text{Shear due to live load on slab (} W_2 l_2 / 2 \text{)} = 0 \times 2.606 / 2$$

$$= 0.000 \text{ kN}$$

$$\text{Total shear force} = 0.000 \text{ kN}$$

$$\text{Factored shear force} = 1.5 * 0.02 \text{ kN}$$

$$= 0.00 \text{ kN}$$

$$\text{Nominal shear stress (TV)} = 0.00 \times 10^3 / (60 \times -44)$$

$$= 0.00 \text{ N/mm}^2$$

$$\text{Permissible Shear stress(Tc)} = 0.29 > 0.00$$

PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

Hence Safe

T_{cmax} As per Table 20 of IS456	=	3.50	N/mm ²
Shear resisted by concrete (V_{uc})	T_{cbd}	=	-0.77 kN
Shear force to resist by reinforcement (V_{us})	$V_{us} = V_u - V_{uc}$	=	0.77 kN
Dia of shear steel	=	8	mm
Spacing of shear stirrups	S_v	=	0.00 mm
Max spacing	=	0.00	mm

B) Design of Side wall

Thickness of side wall	=	150	mm
Height of side wall	=	1240 - 0 + 150	
(Inner depth - ht above FGL+ Base slab thk)	=	1390	mm
Dia of reinforcement bar	=	10	mm
Effective depth ($t_2 - c_s - \phi/2$)	=	150 - 75 - (10 / 2)	
	=	70	mm
Cable Tray angle Fixing Interval	=	1.5	m

Calculation of Loading:

a) Shear & Moment due to weight of cables, supports & man:

Total weight of tray per metre	=	0.90	kN/m
No. of Tray	=	1500	no
Width of Tray supports	=	600	mm
Weight of man with kit	=	1.5	kN/m
Bending Moment due to Cables	=	0.90 x 1500 x 0.6 x (1.5 / 2)	
	$(w \times n \times b' \times (l'/2))$	=	607.50 kNm
Bending Moment due to man	=	1.5 x 0.6	
	=	0.900	kNm

b) Shear & Moment due to Surcharge:

Coefficient of Active Earth pressure	K_a	=	0.42
Surcharge Pressure	=	0	kN/m ²
Horizontal Pressure on side wall due to	=	0.42 x 0	
Surcharge Pressure ($q \times k_a$)	=	0.00	kN/m ²
Moment on side wall	=	$K_a \times q \times H^2 / 2$	



PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ
		=	0.00	kNm
	Shear due to Surcharge	=	$K_a \times q \times h$	
		=	0.00	kN
c)	<u>Shear & Moment due to Soil Pressure:</u>			
	Coefficient of Active Earth pressure K_a	=	0.42	
	Density of soil	=	15.5	kN/m ³
	Pressure due to soil(P)	=	$K_a \times q \times H^2 / 2$	
		=	9.43	kN/m
	Moment due to Soil	=	$P \times h/3$	
		=	6.5	kN/m
	Shear due to soil	=	9.43	kN/m
d)	<u>Shear & Moment due to Water Pressure:</u>			
	Horizontal Base pressure due to water pressure	=	10 x 5	
		=	50	kN/m
	Moment on side wall /1.00 m length	=	$10 \times 1.5 \times 5 \times 5 / 6$	
		=	62.50	kNm
	Shear due to water pressure	=	$50 \times 60 / 1000 \times 1390 / 1000 / 2$	
		=	2.08	kN
e)	<u>Shear & Moment due to pore water pressure: (Triangular water pressure):</u>			
	Coefficient of water pressure	=	-0.36	
	Density of water	=	10	kN/m ³
	Pressure at base due to soil	=	-0.30	kN/m ²
	Shear due to pore water pressure	=	-0.01	kN
	Moment due to pore water pressure	=	-0.01	kNm
	Factored Shear force	=	17.25	kN
	Factored Bending Moment	=	1016.160	kNm
	<u>Main Reinforcement:</u>			
	$M_u / B d^2$	=	3456.327	N/mm ²
	From Table 4, of SP-16			
	Percentage of steel required p_t	=	NaN	%
	Area of steel required	=	NaN	mm ²
	Spacing provided	=	50	mm
	Provide 10 mm dia at 50 mm spacing as Main reinforcement			
	Area of steel provided	=	1570.8mm ² & $p_t = 37.4\%$	
			Hence Unsafe	

PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

Distribution Steel

Dia of bar considered	=	8	mm
Spacing Considered	=	100	mm
Min area of steel required	=	10.80	mm ²
Provide 8 mm dia at 100 mm spacing as distribution steel			
Area of steel provided	=	502.65	mm ²

Hence Safe

Check for Shear:

Factored Shear force	=	17.25	kN
Nominal shear stress (τ_v)	=	$17.25 \times 10^3 / 60 \times 70$	
	=	4.11	N/mm ²
Corresponding τ_c value for the Pt of 0.94	=	0.76 > 4.11	

Hence Unsafe

T _{cmax} As per Table 20 of IS 456	=	3.50	N/mm ²
Shear resisted by concrete (V_{uc})	T _c bd	=	3.19 kN
Shear force to resist by reinforcement (V_{us})	$V_{us} = V_u - V_{uc}$	=	14.06 kN
Dia of shear steel		=	0 mm
Spacing of shear stirrups	S _v	=	0.00 mm
Max spacing		=	0.00 mm
Provide 0 mm dia at 3.50 mm spacing as distribution steel			

C) Design of Base Slab

Thickness of base slab	=	150	mm
Total width of Base slab	=	2900	mm
Wall length considered for design	=	60	mm
Dia of bar considered	=	12	mm
Spacing of reinforcement	=	100	mm

Calculation of Loading:

Live load on cover slab	(1)	=	1.5 kN/m ²
(1.5 x 2900 x 60 / 1000000)		=	0.3 kN
Weight of Cover Slab	(2)	=	(0.006 x 0.06 x 2.606) x 25
		=	0.026 kN
Weight of side walls	(3)	=	(1.24 x 0.15 x 0.06 x 25 x 2)
		=	0.63 kN
Weight of Cables	(4)	=	0.90 kN/m

PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ
Total Weight due to Cables and supports		=	(0.90 x 1500 x 1 x 0.06)	
		=	81.0 kN	
Weight of Base slab (5)		=	(0.15 x 0.06 x 2.9 x 25)	
		=	0.7 kN	
Weight of Man with Kit (6)		=	1.5 kN	
Total Dead load (2+3+5) (L1)		=	1.304 kN	
Total Imposed Load (1+4+6) (L2)		=	82.8 kN	
Total Vertical Load (L1+L2)		=	84.065 kN	
Checking Bearing Pressure				
Maximum Bearing Pressure=Total Vertical Load/Area of Base slab		=	483.13>67.6 kN/m ²	
Hence Unsafe				
 <u>Main reinforcement for Base slab:</u>				
Factored Gross pressure 1.5 * 483.13		=	724.695 kN/m ²	
Net upward pressure		=	724.695 - (150 / 1000 * 30)	
		=	720.195 kN/m ²	
Max bending at ends		=	720.195 x (2600 / 1000 + 150 / 1000) ² / 12	
		=	453.87 kNm	
Max moment transferred from side wall		=	1016.160 / (60 / 1000)	
		=	16936.00 kNm	
Total Moment at ends		=	453.87 + 16936.00	
		=	17389.87 kNm	
Total moment at centre		=	453.87 kNm	
Design Moment		=	17389.87 kNm	
Effective depth		=	150 - 50-12 / 2	
		=	94 mm	
Mu/Bd ²		=	(17389.87 x 10 ^ 6) / (60 x 94 x 94)	
		=	32801.17 N/mm ²	
From Table4 of SP 16				
Percentage of steel required pt		=	NaN	
Area of steel required		=	NaN * 94 * 60 / 100	
		=	NaN mm ²	
Spacing of Main reinforcement		=	100 mm	



PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

Therefore Provide 12 mm dia at 100 mm spacing as Main reinforcement at bottom

Area of steel provided = 1130.973 > NaN

Hence Unsafe

Max allowable spacing for reinforcement = 282.00 > 100 mm

Hence Safe

Distribution steel for Base slab:

Dia of bar = 12 mm

Spacing of bar = 100 mm

Min area of steel required (0.12 bd/100) = 11 mm²

Therefore Provide 12 mm dia at 100 mm spacing as distribution reinforcement at bottom

Area of steel provided = 67.86 > 11 mm²

Hence Safe

Max. allowable Spacing of 5 * 94 = 470 mm

Reinforcement bar as per CL.26.3.3(b) of or 450 mm

IS 456 -lesser of = 450 > 100 mm

Hence Safe

Top reinforcement (for Base slab)

Dia of reinforcement bar = 20 mm

Effective depth 150 - 50 - 20 / 2 = 90 mm

Spacing = 1200 mm

$M_u / B d^2 (453.87 \times 1.5 \times 10^6) / (60 \times 90 \times 90)$ = 1400.83

From Table 4 of SP16

Percentage of steel required pt = NaN

Minimum percentage of steel as per IS-456 = 0.12

Area of steel required MAX(0.12 + NaN)x 90 x 60 / 100 = NaN mm²

Therefore Provide 20 mm dia at 1200 mm spacing as Main reinforcement at Top

Area of reinforcement provided (20² x 3.14 / 4 x 1000 / 1200) = 261.80 > NaN mm²

Hence Unsafe

Distribution steel required (0.12bd/100) = 10.80 mm²

Therefore provide 8mm dia at 200mm C/C as distribution reinforcement

Check for Shear:



PROJECT	NTPC REL - 1200MW BoS SPV Plant, Khavda Solar Park, GJ	Document No	Date	Rev No
		01cantfilsep-01cantfilsep	15-6-2026	1
TITLE	Cable Trench IS	DESIGNED	CHECKED	APPROVED
		PMT/DMV	TJK	VRJ

Factored Shear force	=	126.0977	kN
Nominal shear stress (τ_v)	=	$126.0977 \times 10^3 / 60 \times 94$	
	=	22.36	N/mm ²
Corresponding τ_c value for the pt of	=	$0.76 > 22.36$ N/mm ²	
Hence Unsafe			
τ_{cmax}	=	3.50	N/mm ²
Shear resisted by concrete (V_{uc})	τ_{cbd}	=	4.29 kN
Shear force to resist by reinforcement (V_{us})	$V_{us} = V_u - V_{uc}$	=	121.81 kN
Dia of shear steel	=	8	mm
Spacing of shear stirrups	S_v	=	0.00 mm
Max spacing	=	0.00	mm

Structural Design of Angle Support for Cable Trench:

Size of support angle - 650 wd	=	ISA 50x50x6	
Section modulus 'Z _{xx} '	=	3.6	cu.cm
Load Calculation:			
Total Weight /m width	=	0.90	kN/m
Support fixing Interval	=	1.5	m
Total Length of support	=	0.60	m

Fixing arrangement detail :

The cable support angle (ISA 50x50x6) is welded to the insert plate and the insert plate is embedded into the side wall with suitable lugs.

Hence, effective overhang	=	0.5	m
Maximum moment at support	=	$0.90 \times 1.5 \times 0.60 / 2$	
	=	0.37	Nm
Section modulus Required	=	$0.37 \times 100 / 0.66 \times 2548$	
	=	0.22	cu.cm

Hence Ok